

TRAVELOS — CORPORATE TRAVEL MANAGEMENT SYSTEM

End-to-End System Flow, Technical Wireframes & Architecture Blueprint

STAKEHOLDERS

1

Traveling Employee

Raise · Track · Upload · Close

2

Approving Manager

Review · Remark · Decision

3

Travel Team (Desk)

Bookings · Timeline · Assets

4

Financial Team

Audit Claims · Credit Sal

5

Risk & Admin

Policy · SOS · Governance

SYSTEM FLOW (END-TO-END LIFECYCLE)

Phase 1: Landing & Login Screen

Auth Module

[Emp Dash Login Screen]

ID: EMP001 / EMP002

PW:

Log In → JWT Bearer

[Elevated Role Cards]

- Manager: Passcode 111111
 - Travel Desk: Passcode 222222
 - Finance: Passcode 333333
 - Admin: Passcode 444444
- * Instant role routing without ID input.

Phase 2: Employee Dashboard

Employee View

[Profile Summary]

- Name: Arjun Sharma
- DOJ / Dept: Active (Engg)
- Govt ID: Verified
- Docs: Attached

[Trip Registry]

+ Raise Travel Request

Active Trips	Closed Trips	Upcoming
TRIP-1024 (Live)	TRIP-0982 (Done)	TRIP-1055 (Pending)

Access live tracking, voucher downloads & closure triggers.

Phase 3: Create & Submit Request

Form Input

[Raise Travel Request Form]

- Project & Client: Project #8492 · Client #C-901
- Dest & Duration: Mumbai → London (7 Days)
- Needs: ☒ Flight ☒ Hotel ☒ Cab ☒ Extra Luggage
- Asset Allocation: ThinkPad X1 + Testing Device
- Estimated Cost: Total Budget: \$3,450

[Submit Request] → PENDING_AUTO_VAL

Phase 4: Automatic Validation

5-Check Engine

[AUTOMATIC VALIDATION]

- ✓ Is Emp Active?
- ✓ Is Valid Client?
- ✓ Mandatory Details & Company Policy OK?
- ✓ Does Project Exist?
- ✓ Budget Available?

✓ Yes → PENDING_MGR

X No → REJECTED_SYS

Phase 5: Manager Review

Manager

[Manager Dashboard Queue]

Pending: 3 | Approved: 14 | Rejected: 1

- Audit Review: Business purpose & budget fit
- Asset Check: Verify hardware allocation
- Remarks: Mandatory feedback input

[APPROVE]

[REJECT]

Status updates to APPROVED on sign-off.

Phase 6: Travel Desk & Booking

Travel Desk

[Travel Team (Travel Desk) Execution]

- Flight Booking: PNR: AI-204 · Dep: 08:00 AM · Arr: 11:00 AM
- Cab Assignment: Cab: DL-01-AB-1234 (Ramesh Kumar)
- Hotel Reservation: Grand Hyatt (Booking ID: HBK-90234)
- Asset Handover: HW-Asset-8492 logged & dispatched
- Checklist Timeline: Pre-fills & schedules all 7 milestone times

Activate Itinerary → Status: [ACTIVE]

Phase 7: Active Trip & Live Tracking

Live Verification

[7-Milestone Timeline with Mandatory Verification]

1	2	3	4	5	6	7
Flight Board	Flight Land	Cab Pickup	Hotel In	Hotel Out	Return Fit	Journey End
Seat No	Terminal	OTP/Code	Book ID	Pending	Pending	Pending

* Milestone accuracy: On-Time / Delayed / Early tracked vs Travel Desk schedule.

Phase 8: Financial Expense Settlement

Finance (Accounts)

[Financial Team (Accounts) Review]

- Upload Bills: Employee uploads receipts (Meals, Cabs, Emergency).
- Audit & Verification: Accounts team reviews claim categories & bills.
- Salary Reimbursement: Approved amount is CREDITED to employee salary.

* Approved expenses transition to status 'CREDITED' prior to trip closure.

Phase 9: Trip Closure & Reconciliation

4-Condition Gate

[Close Trip Automated Verification]

- ① Trip Journey Ended: Milestone 7 verified with arrival proof.
- ② Assets Returned: Physical hardware logged as returned.
- ③ Expenses Settled: All expense claims in CREDITED state.
- ④ Dates & Policy: Current date is past scheduled end date.

All 4 Verified → Finalize

[CLOSE TRIP] → Status: CLOSED

Frontend: Angular 16+ · Glassmorphism · RxJS

Backend: Spring Boot 3.2 · Java 17+ · Spring Security

Security: JWT Bearer · Role Guards

Database: H2 File-based · Hibernate JPA

Specs: 24 Endpoints · 8 Entities